

Daily Review

Lesson 1

- 1 Name the four types of engineer.
- 2 Who designs a bridge's foundations?
- 3 What is a prototype?
- 4 Order: build, improve, design, test
- 5 A chemical engineer's job, and why?

Electrical, mechanical,
civil, chemical

Civil

A first working version,
built to test

Design, build, test,
improve

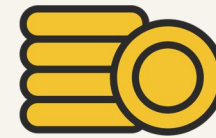
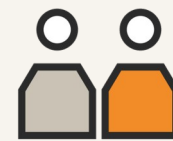
A coating — so steel
won't rust

TEST 1 REVISION • LESSON 2 OF 4

Impacts of Engineering

Who gains, who loses — and what happens to the planet?

S3 Engineering Science



THE BIG QUESTION

The Borders Railway reopened in 2015. Who gained — and did anyone lose?



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PART 1

Impacts of Engineering

Good and bad effects on people, money and the natural world.

By the end of this part...

- 1 I can describe positive and negative effects on society
- 2 I can describe positive and negative economic effects
- 3 I can describe positive and negative environmental effects
- 4 I can use real examples in my answers
- 5 I can explain my ideas clearly using what + why

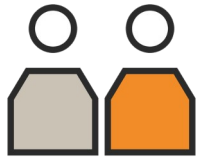
SECTION A

Three Kinds of Impact

More jobs, cleaner air, safer streets — the same kind of effect?

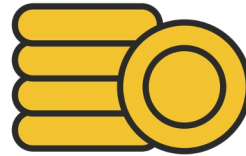
The Three Types of Impact

Impact = an effect or change caused by something



Social

people's lives:
safety, travel,
health



Economic

money and jobs:
costs, profits,
tourism



Environmental

the natural world:
pollution, CO₂,
habitats, waste

Worked Example — Sort the Borders Railway

Trap: “saves money” is always economic

Social

easier trips to work or college in Edinburgh

Economic

tourists spend money in Borders shops

Environmental

fewer car trips, so less CO₂



Show Me — Which Type of Impact?

SHOW ME BOARDS

- 1 Faster broadband lets pupils do homework online.
- 2 A new ski centre brings tourists who spend in local shops.
- 3 A new reservoir floods a valley where deer lived.
- 4 LED street lights cut the council's electricity bill.
- 5 Electric cars give out no exhaust in a town centre.

Social

Economic

Environmental

Economic

Environmental

Show Me – Which Type of Impact?

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Self-checkouts mean fewer checkout jobs.
- 2 A new footbridge: children no longer cross a busy road.
- 3 Old phones thrown away fill up landfill.
- 4 The LED lights also mean less electricity from power stations.
- 5 Why can one project have all three types of effect?

Economic

Social

Environmental

Environmental

It changes people,
money and nature

SECTION B

Good and Bad

Is a wind farm good or bad? It depends who you ask.

Both Sides — What, Why and Who

Most projects have positive AND negative effects

What

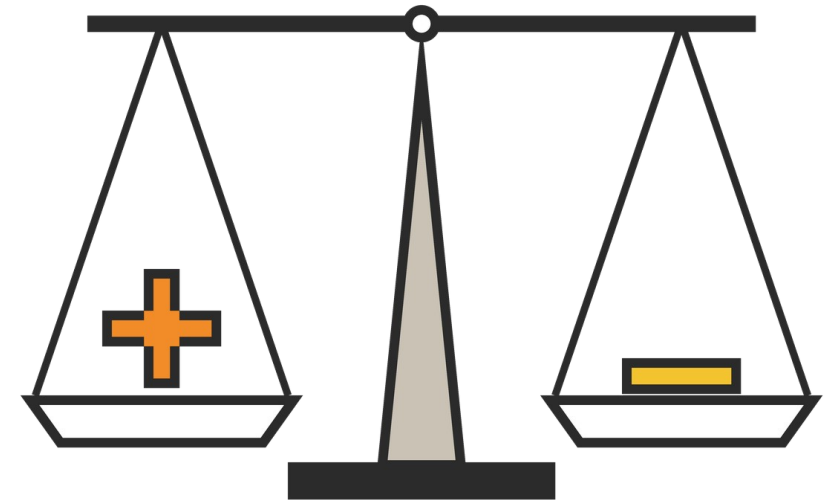
the effect itself

Why

why it happens, or why it matters

Who

who or what is affected



Worked Example – Smartphones

Describe one positive and one negative social effect. (2 marks)

Positive

people can call for help from anywhere, so they are safer when out alone

Negative

some people use their phones late at night, so they get less sleep

Scores 0

“they're useful” – no what, no why



Show Me — Self-Checkouts

SHOW ME BOARDS

- 1 Positive economic effect for the supermarket
- 2 Negative economic effect for staff
- 3 Positive social effect for shoppers
- 4 Negative social effect for shoppers
- 5 Which scores: A “bad for jobs” or B “fewer till staff, so some lose jobs”?

Pays fewer staff — so saves money

Fewer till jobs — so some lose work

Shorter queues — so a quicker trip

Some find screens hard — so they need help

B — it says what AND why

Show Me — Online Gaming

SHOW ME BOARDS

EXTRA PRACTICE

- 1 A positive social effect
- 2 A negative social effect
- 3 An economic effect for a games company
- 4 An environmental effect when old consoles are thrown away
- 5 Improve: “gaming is bad for you”.

Play with friends far away — so less lonely

Hours online — so less sleep or exercise

Sells games — so makes a profit

Electronic waste in landfill

Late-night gaming means less sleep

SECTION C

Real Examples

“A wind farm is good for the environment.” Which one? Where?

Four Scottish Projects (1 of 2)

Real examples you can use in an answer



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Whitelee Wind Farm

electricity with no fuel burned,
so less CO₂



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Pitlochry Dam

renewable electricity – but it
blocks salmon

Four Scottish Projects (2 of 2)

Real examples you can use in an answer



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Queensferry Crossing

wind shields keep it open, so
journeys are reliable



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Borders Railway

easier travel to Edinburgh;
tourists visit

Worked Example – Build Up an Answer

Describe one negative environmental effect of a hydro dam.

Scores 0

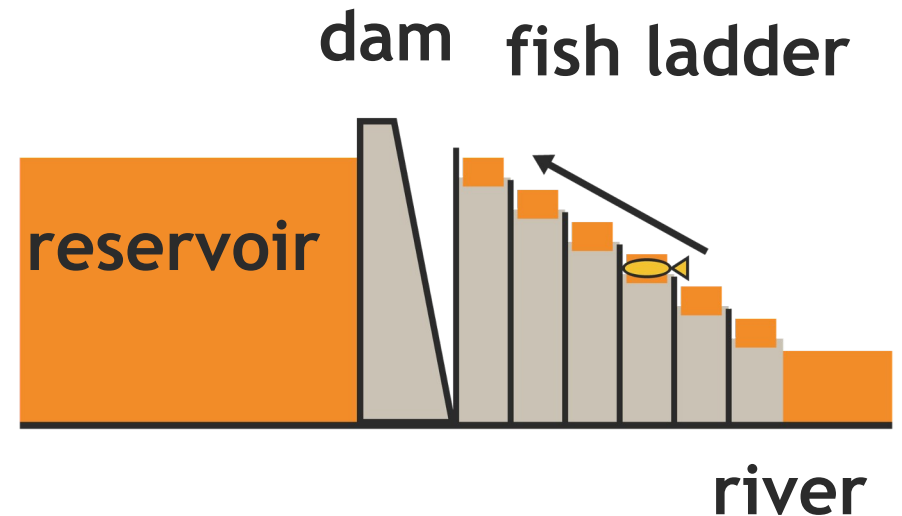
“dams are bad for fish”

What + why

the dam blocks the river, so salmon can't swim up to breed

+ real example

Pitlochry Dam has a fish ladder – a staircase of pools salmon can climb



Show Me — Real Examples

SHOW ME BOARDS

- 1 Name one real renewable-energy project in Scotland.
- 2 Queensferry Crossing: one social effect, what + why
- 3 Whitelee: one positive environmental effect, what + why
- 4 Why does Pitlochry Dam have a fish ladder?
- 5 A real engineering project near you — and one impact.

Whitelee Wind Farm,
Pitlochry Dam

Stays open in wind — so
people get to work

No fuel burned — less
 CO_2

So salmon can get past
the dam to breed

Your own answer

Show Me — Real Examples

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Borders Railway: who gains money, and why?
- 2 One negative effect of building the Borders Railway
- 3 Improve with a real example: “wind farms help the environment”.
- 4 Can one effect be good for one group and bad for another?
- 5 Why does naming a real project make an answer stronger?

Borders shops — tourists come by train

It cost a lot to build

Whitelee burns no fuel,
so less CO₂

Yes — self-checkouts:
shop gains, staff lose

It proves the effect
really happens

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

“A new bus route means people can get to the hospital more easily.” Which type?

- 1 economic
- 2 social
- 3 environmental
- 4 it isn't an impact

Answer 2

It changes people's lives – getting to a service is social.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

“More tourists visit, so local cafés take more money.” Which type?

- 1 social
- 2 environmental
- 3 economic
- 4 all three equally

Answer 3

The effect is money coming in – that is economic.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

Which is an environmental effect of solar panels on a school roof?

- 1 the school saves money on bills
- 2 pupils learn about energy
- 3 the panels cost a lot to fit
- 4 less fuel burned, so less CO₂

Answer 4

Saving money is economic, even on a green project.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

Can one project have both positive and negative effects?

- 1 yes – most projects have both
- 2 no – it's either good or bad
- 3 only if something goes wrong
- 4 only very large projects

Answer 1

Almost every project helps someone and costs something.

Finger Vote – Common Mix-ups

FINGER VOTE

Think first. Fingers on your chest – show on 3, 2, 1.

“Describe one negative environmental effect of a new wind farm.”
Which answer gets the mark?

- 1 “it's bad for the environment”
- 2 “it costs a lot to build”
- 3 “birds can be killed by the blades”
- 4 “wind turbines”

Answer 3

Only 3 names a real effect on nature. 2 is economic.

Mini Test 2

On your own, in silence, pen only.
10 minutes – 8 marks.
Then swap and mark from the next slides.

Mini Test 2 – Impacts of Engineering

Mini Test 2 – Marking

ANSWERS

Swap sheets. Tick or cross, then total out of 8.

Q1 (1 + 1)

+ quieter, no exhaust, so cleaner air for people. – fewer routes while buses recharge, or disruption.

A money answer scores 0 in Q1 – money is Q2.

Q2 (1 + 1)

+ electricity costs less than diesel, or jobs fitting chargers. – new buses and chargers are expensive, or diesel mechanics lose work.

Mini Test 2 – Marking

ANSWERS

“Better for the environment” on its own scores 0.

Q3(a) (1)

No exhaust from the buses, so less air pollution in town.

Q3(b) (1)

Mining metals for batteries damages land, or old buses and batteries become waste.

Q4 (1 + 1)

A named real project with a true impact (e.g. Whitelee: no fuel burned, so less CO₂) + the right type.

Plenary

SHOW ME BOARDS

1 What does an economic impact affect?

Money and jobs

2 One negative environmental effect of building work.

Habitats damaged, or waste

3 One positive and one negative social effect of smartphones.

Safer when out – but less sleep

4 A real project and one impact, what + why.

Whitelee – no fuel burned, so less CO₂

5 The three things a strong impact answer says.

What, why and who

Plenary

SHOW ME BOARDS

EXTRA PRACTICE

1 “Staff lose their jobs” – which type?

Economic

2 Improve: “it's good for the environment”.

Say what changes in nature, and why

3 What problem does a fish ladder solve?

Salmon can't pass the dam

4 Which type of impact do you find hardest?

Your own answer

5 Which I Can are you least sure of?

Your own answer

LOOKING AHEAD

Lesson 3 — Energy

Forms of energy, transfers, and useful and wasted energy.